

DATA MINING-IV

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Clustering:

Clustering is the process of grouping a set of data objects into multiple groups or clusters so that objects within a cluster have high similarity, but are very dissimilar to objects in other clusters.

Dissimilarities and similarities are assessed based on the attribute values describing the objects and often involve distance measures.¹

Clustering as a data mining tool has its roots in many application areas such as biology, security, business intelligence, and Web search.

What Is Cluster Analysis?

Cluster analysis or simply clustering is the process of partitioning a set of data objects (or observations) into subsets.

Each subset is a cluster, such that objects in a cluster are similar to one another, yet dissimilar to objects in other clusters.

The set of clusters resulting from a cluster analysis can be referred to as a clustering.

Cluster analysis has been widely used in many applications such as business intelligence, image pattern recognition, Web search, biology, and security.

In business intelligence, clustering can be used to organize a large number of customers into groups, where customers within a group share strong similar characteristics.

In image recognition, clustering can be used to discover clusters or “subclasses” in handwritten character recognition systems

a keyword search may often return a very large number of hits (i.e., pages relevant to the search) due to the extremely large number of web pages.

Clustering can be used to organize the search results into groups and present the results in a concise and easily accessible way.

clustering is sometimes called automatic classification.

Clustering is also called data segmentation in some applications because clustering partitions large data sets into groups according to their similarity.

Clustering can also be used for outlier detection,

Applications of outlier detection include the detection of credit card fraud and the monitoring of criminal activities in electronic commerce.

Cluster analysis tools based on k-means, k-medoids, and several other methods also have been built into many statistical analysis software packages or systems, such as S-Plus, SPSS, and SAS.

In machine learning, recall that classification is known as supervised learning because the class label information is given, that is, the learning algorithm is supervised in that it is told the class membership of each training tuple.

Clustering is known as unsupervised learning because the class label information is not present. For this reason, clustering is a form of learning by observation, rather than learning by examples.

Requirements for Cluster Analysis:

1. Scalability
2. Ability to deal with different types of attributes:
3. Discovery of clusters with arbitrary shape
4. Requirements for domain knowledge to determine input parameters
5. Ability to deal with noisy data
6. Incremental clustering and insensitivity to input order:
7. Capability of clustering high-dimensionality data
8. Constraint-based clustering
9. Interpretability and usability
10. The partitioning criteria:
11. Separation of clusters
12. Similarity measure:
13. Clustering space

Partitioning methods:

Given a set of n objects, a partitioning method constructs k partitions of the data, where each partition represents a cluster and $k \leq n$. it divides the data into k groups such that each group must contain at least one object.

partitioning methods conduct one-level partitioning on data sets

The basic partitioning methods typically adopt exclusive cluster separation. That is, each object must belong to exactly one group.

Most partitioning methods are distance-based

Given k , the number of partitions to construct, a partitioning method creates an initial partitioning.

It then uses an iterative relocation technique that attempts to improve the partitioning by moving objects from one group to another.

The clusters are formed to optimize an objective partitioning criterion, such as a dissimilarity function based on distance, so that the objects within a cluster are “similar” to one another and “dissimilar” to objects in other clusters in terms of the data set attributes.

Most applications adopt popular heuristic methods, such as greedy approaches like the k-means and the k-medoids algorithms,

- Find mutually exclusive clusters of spherical shape
- Distance-based
- May use mean or medoid(etc.) to represent cluster center.
- Effective for small- to medium-size data sets

k-Means: A Centroid-Based Technique:

“How does the k-means algorithm work?”

The k-means algorithm defines the centroid of a cluster as the mean value of the points within the cluster.

It proceeds as follows.

First, it randomly selects k of the objects in D , each of which initially represents a cluster mean or center.

For each of the remaining objects, an object is assigned to the cluster to which it is the most similar, based on the Euclidean distance between the object and the cluster mean.

The k-means algorithm then iteratively improves the within-cluster variation. For each cluster, it computes the new mean using the objects assigned to the cluster in the previous iteration.

All the objects are then reassigned using the updated means as the new cluster centers.

The iterations continue until the assignment is stable, that is, the clusters formed in the current round are the same as those formed in the previous round.

Algorithm: k-means. The k -means algorithm for partitioning, where each cluster’s center is represented by the mean value of the objects in the cluster.

Input:

- k : the number of clusters,
- D : a data set containing n objects.

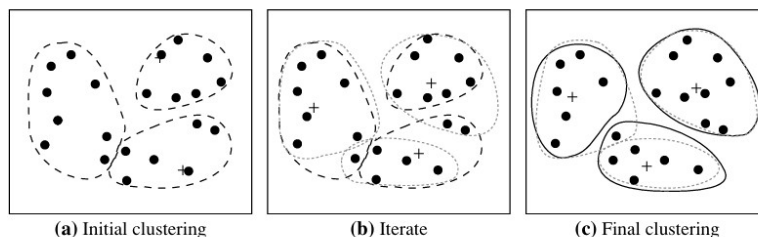
Output: A set of k clusters.

Method:

- (1) arbitrarily choose k objects from D as the initial cluster centers;
- (2) **repeat**
- (3) (re)assign each object to the cluster to which the object is the most similar, based on the mean value of the objects in the cluster;
- (4) update the cluster means, that is, calculate the mean value of the objects for each cluster;
- (5) **until** no change;

2 The k-means partitioning algorithm.

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Clustering of a set of objects using the k -means method; for (b) update cluster centers and reassign objects accordingly (the mean of each cluster is marked by a +).

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Hierarchical methods:

A hierarchical method creates a hierarchical decomposition of the given set of data objects.

A hierarchical method can be classified as being either agglomerative or divisive, based on how the hierarchical decomposition is formed.

The agglomerative approach, also called the bottom-up approach, starts with each object forming a separate group. It successively merges the objects or groups close to

one another, until all the groups are merged into one (the topmost level of the hierarchy), or a termination condition holds.

The divisive approach, also called the top-down approach, starts with all the objects in the same cluster. In each successive iteration, a cluster is split into smaller clusters, until eventually each object is in one cluster, or a termination condition holds.

Hierarchical clustering methods can be distance-based or density- and continuity based.

Density-based methods:

Most partitioning methods cluster objects based on the distance between objects. Such methods can find only spherical-shaped clusters and encounter difficulty in discovering clusters of arbitrary shapes.

Other clustering methods have been developed based on the notion of density. Their general idea is to continue growing a given cluster as long as the density (number of objects or data points) in the “neighborhood” exceeds some threshold.

For example, for each data point within a given cluster, the neighborhood of a given radius has to contain at least a minimum number of points. Such a method can be used to filter out noise or outliers and discover clusters of arbitrary shape.

Density-based methods can divide a set of objects into multiple exclusive clusters, or a hierarchy of clusters. Typically, density-based methods consider exclusive clusters only, and do not consider fuzzy clusters.

Grid-based methods:

Grid-based methods quantize the object space into a finite number of cells that form a grid structure. All the clustering operations are performed on the grid structure. The main advantage of this approach is its fast processing time, which is typically independent of the number of data objects and dependent only on the number of cells in each dimension in the quantized space.

Method	General Characteristics
Partitioning methods	– Find mutually exclusive clusters of spherical shape – Distance-based – May use mean or medoid (etc.) to represent cluster center – Effective for small- to medium-size data sets
Hierarchical methods	– Clustering is a hierarchical decomposition (i.e., multiple levels) – Cannot correct erroneous merges or splits – May incorporate other techniques like microclustering or consider object “linkages”
Density-based methods	– Can find arbitrarily shaped clusters – Clusters are dense regions of objects in space that are separated by low-density regions – Cluster density: Each point must have a minimum number of points within its “neighborhood” – May filter out outliers
Grid-based methods	– Use a multiresolution grid data structure – Fast processing time (typically independent of the number of data objects, yet dependent on grid size)

0.1 Overview of clustering methods discussed in this chapter. Note that some algorithms may combine various methods.

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Density-Based Methods:

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Partitioning and hierarchical methods are designed to find spherical-shaped clusters. They have difficulty finding clusters of arbitrary shape such as the “S” shape and oval clusters

To find clusters of arbitrary shape, alternatively, we can model clusters as dense regions in the data space, separated by sparse regions.

This is the main strategy behind density-based clustering methods, which can discover clusters of nonspherical shape

In this section, you will learn the basic techniques of density-based clustering by studying three representative methods, namely, DBSCAN (Section 10.4.1), OPTICS (Section 10.4.2), and DENCLUE (Section 10.4.3)

DBSCAN: Density-Based Clustering Based on Connected Regions with High Density:

“How can we find dense regions in density-based clustering?”

The density of an object o can be measured by the number of objects close to o .

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) finds core objects, that is, objects that have dense neighborhoods. It connects core objects and their neighborhoods to form dense regions as clusters.